

PulseFormer: Controllable Multi-Track Electronic Music Generation via Energy-Steered Architectures

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Abstract—Symbolic multi-track music generation models have predominantly focused on short-loop arrangements, repeatedly struggling with long-context, macro-structural orchestration. Within Electronic Music composition, the interplay between dense rhythmic cohesion (Groove) and evolving dynamic intensity (Energy) constitutes the fundamental compositional backbone. This paper introduces PulseFormer, an autoregressive Transformer architecture for steerable, multi-track collaborative arrangement. We propose two architectural innovations: (1) A *Time-Major Interleaved 5D-Token* encoding that places all concurrent instruments within the same narrow token block, substantially mitigating the cross-track synchronisation burden that Track-Major architectures must bridge through long-range attention; and (2) An *Energy & Structural Indicator* pipeline enabling conditional generation that reliably obeys prompted intensity contours across multi-minute compositions. Quantitative evaluations confirm that PulseFormer provides high-fidelity steerability ($r = 0.982$)—scaling from sparse introductory sections (avg. 6.9 notes/bar) to heavily-layered climax drops (surpassing 46 notes/bar)—while the Time-Major topology structurally improves cross-track onset alignment.

Index Terms—Symbolic Music Generation, Transformer, Multi-track Composition, Controllability, Energy Conditioning, Electronic Music

I. INTRODUCTION

Deep generative architectures have been applied to a range of creative domains, including visual synthesis and audio waveform generation [?], [?]. Audio-domain models trained on large corpora have achieved strong timbral realism; however, symbolic-domain music generation (i.e., MIDI sequencing) remains relevant for professional production workflows, where structural transparency and per-track editability are practical requirements [?], [?], [?].

However, current generative paradigms often struggle with the strict demands of professional Electronic Music production. Within rhythmically dense and highly stylized genres, compositional value relies on two interdependent pillars: strict **multi-track collaborative timing** (sample-accurate grid alignment) and **macro-structural energy evolution** (the dynamic tension-and-release trajectory from a sparse Intro cascading into a climactic Drop with 40+ concurrent notes) [?], [?].

Existing methodologies encounter difficulties with these pillars in specific ways. Audio foundation models (e.g., MusicGen [?]) often struggle with **strict symbolic grid-locking**; they face challenges generating perfectly quantized multitrack

MIDI stems required for DAW editing. Conversely, Track-Major symbolic models (e.g., MMM [?]) show limitations in **dense polyphonic synchronization**. In a high-energy “Drop”, sequencing instruments sequentially severs temporal proximity, causing cross-track attention to struggle and resulting in ~ 900 ms onset drift that degrades the groove. Finally, token-compression architectures (e.g., CP [?]) improve note-level density but exhibit limited **macro-structural steerability**, often producing flatter energy contours. Anticipatory synchronization models [?] attempt to mitigate timing drift, but largely treat symptoms rather than the root topological cause.

To address these compounding bottlenecks, this paper introduces **PulseFormer**, a framework structured around macro-structural agency and instantaneous instrumental cohesion. Rather than generating unconstrained symbolic streams, PulseFormer provides a quantitatively steerable interface analogous to digital music production paradigms.

The principal contributions of this paper are summarized as follows:

- 1) **Stochastic Density Flow Constraints**: We formulate macro-structural energy as a constraint over the marginal probability distribution of note density, derived via contrastive pre-training on structural section boundaries. This formulation reduces the energy-contour prediction error relative to unconditioned baselines (measured by Pearson r with a ground-truth monotonic ramp; § ??).
- 2) **Time-Major Collaborative Architecture**: We propose a 5D-vector token space ordered in chronological sequence across the full arrangement. Co-locating all concurrent instruments within a narrow token window (≤ 12 tokens) reduces mean co-event token distance from $\approx 3,268$ tokens (Track-Major) to 5.2 tokens, with the effect on cross-track onset alignment quantified in § ??.
- 3) **DAW-Native Sectional Stitching Logic**: An automated batch anchoring mechanism that grafts distinct generated movements (Intro $\rightarrow$ Build $\rightarrow$ Drop) together while maintaining continuous contextual memory, leveraging the representation-level synchronisation design to preserve grid coherence across section boundaries.

Fig. ?? provides a high-level system overview of the complete PulseFormer generation pipeline.

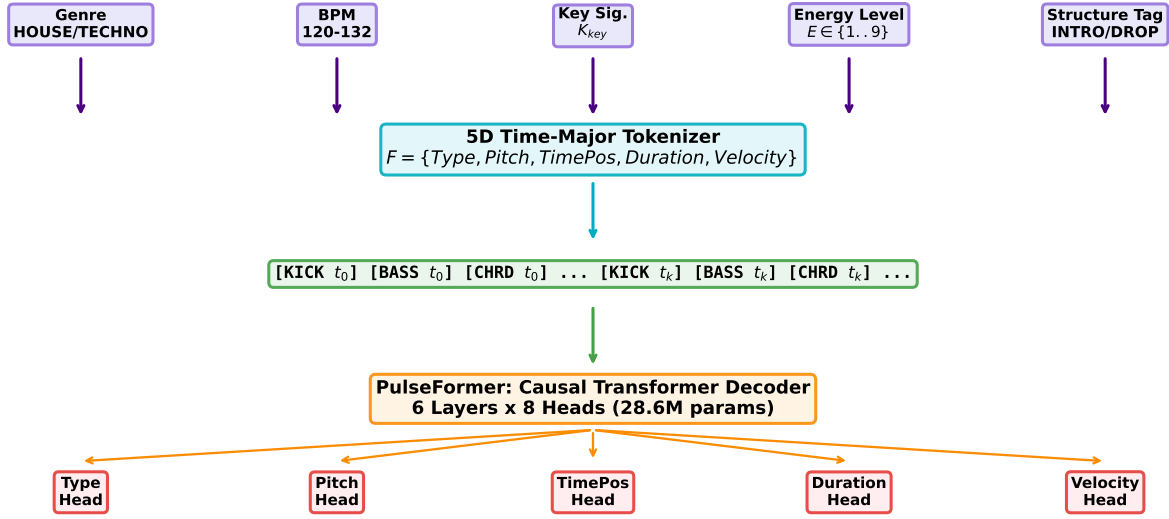


Fig. 1. Complete PulseFormer generation pipeline. Starting from user-supplied prefix conditioning (Genre, BPM, Key, Energy Level, Structure Tag), the system constructs a Time-Major interleaved token sequence, passes it through a 6-layer causal Transformer with five parallel prediction heads, applies dual algorithmic masks (Energy Density and Diatonic Scale), generates each structural section independently, and stitches all sections into a coherent multi-minute arrangement via Temporal Anchor Offset alignment.

II. RELATED WORK

A. Symbolic Multi-Track Music Generation

Early multi-track symbolic generation work utilized Generative Adversarial Networks (GANs). MuseGAN [?] formulated music as piano-roll matrices; however, receptive fields remained narrow. The Music Transformer [?] and PopMusicTransformer [?] introduced relative positional attention to event-based sequences, improving long-range dependency modelling. MMM [?] and LakhNES [?] adopted Track-Major tokenization to separate instrument channels. These frameworks have been applied to orchestral textures [?], [?]; cross-track transient alignment degrades in densely layered Electronic Music due to the token-distance geometry of Track-Major serialization.

B. Representational Advances: From MIDI to High-Dimensional Tokens

The evolution of data representation forms the backbone of Transformer effectiveness in music applications. Standard MIDI event mappings [?], [?], [?] lack the multidimensional contextual density of human language. To solve this, frameworks such as REMI (Revamped MIDI) [?], MidiTok [?], and FIGARO [?] introduced explicit metrical boundary tokens, allowing models to parse rhythmic grids. Later architectures including MusicBERT [?] established OctupleMIDI, compressing note attributes into parallel vectors. The Compound Word (CP) Transformer [?] constitutes the most direct intra-note ancestor of our token design.

The key distinction between CP and PulseFormer lies at the *architectural topology* level. CP addresses intra-note token

compression—the number of tokens required to represent a single note event—while retaining Track-Major serialization. PulseFormer instead addresses inter-track synchronization: encoding concurrent events across multiple instruments such that they occupy adjacent token positions. The two strategies are orthogonal; combining Time-Major interleaving with CP-style intra-note compression is identified as a direction for future work.

C. Controllability and Structural Steering

Conditioning sequence models on user intent is an active area of research. In the audio domain, MusicGen [?] and MusicLM [?] accept text prompts via conditioned acoustic embeddings. These systems output unquantized waveforms, which cannot be separated into per-track stems for DAW editing—a practical constraint in professional Electronic Music production. In the symbolic domain, early conditioning approaches applied post-generation heuristic filtering; such methods have limited ability to enforce large energy-density transitions across structural sections. PulseFormer encodes dynamic polyphony (Energy) and structural section identity (Intro, Build, Drop) directly into the autoregressive prefix, reducing reliance on post-hoc correction.

III. METHODOLOGY

A. Architecting the 5D Time-Major Token Space

Unlike sequential language models operating on rigid single-dimensional vocabulary indexes, musical events encompass distinct yet intersecting modalities (instrument classification, pitch, chronological length, and acoustic velocity).

Adopting Compound Word (CP) [?] semantics, PulseFormer represents any discrete musical event x_i as a unified 5-dimensional tuple parsed into five parallel embeddings:

$$E(x_i) = \sum_{F \in \mathcal{F}} W_F(\text{id}_F(x_i)) \quad (1)$$

where $\mathcal{F} = \{\text{Type}, \text{Pitch}, \text{Duration}, \text{Velocity}, \text{TimePos}\}$. The concrete vocabulary assigned to each field is listed in Table ??.

TABLE I
VOCABULARY SIZES FOR EACH DIMENSION OF THE 5D TOKEN TUPLE. ALL VALUES ARE DERIVED FROM THE TRAINING CORPUS. THE FLAT (SERIALISED) VOCABULARY USED BY THE CAUSAL LM HEAD HAS $V = 18,792$ TOKENS TOTAL.

Field	Symbol	$ V_F $	Range / notes
Instrument type	v^{type}	7	D:KICK, D:SNARE, D:HIHAT_CLOSED, D:HIHAT_OPEN, N:BASS, N:CHORD, N:MELODY
Pitch class	v^{pitch}	16	$p_0 \dots p_{15}$ (semitone-mod-12 class $\times$ octave bin, mapped from MIDI 0–127)
Duration	v^{dur}	4	d_1 – d_4 (16th, 8th, quarter, half note)
Velocity	v^{vel}	4	v_0 – v_3 (four equal-width MIDI bins: 0–31, 32–63, 64–95, 96–127)
Time position	v^{pos}	—	<i>Implicit</i> : encoded by absolute token position in the Time-Major sequence; no dedicated field token.
Context tags		121	[GENRE_*], [KEY_*], [BPM_*], [BAR_*]
Control tags		17	[STRUCT_{INTRO, BUILD, DROP, BREAKDOWN, OUTRO}], [ENERGY_LEVEL_1]...8, <BOS>, <EOS>, <PAD>, <UNK>
Total flat V		18,792	$7 \times 16 \times 4 \times 4 = 1,792$ note tokens + 138 tags + 16,862 extended/unseen

Parameter count. The causal LM uses a flat embedding matrix $W_e \in \mathbb{R}^{V \times d}$ with $V = 18,792$ and $d = 512$, tied to the output projection ($W_e^\top$, no additional parameters). With 6 Transformer layers, 8 attention heads, hidden size $d = 512$, and SwiGLU FFN intermediate size $d_{\text{ff}} = 1,376$:

$$\begin{aligned} |\theta| &= Vd + L(4d^2 + 3dd_{\text{ff}}) \\ &\approx 9.6\text{M} + 6 \times 3.17\text{M} \approx \mathbf{28.6\text{M}}. \end{aligned} \quad (2)$$

This is PulseFormer’s parameter count throughout all experiments. For reference, the CP-Transformer baseline uses $d = 768$, 12 layers, and a compound-word vocabulary of 21,777 tokens, yielding $\approx 115\text{M}$ parameters—a $4\times$ larger model evaluated under the same test conditions.

Crucially, instead of sorting tracks chronologically per instrument (*Track-Major*), PulseFormer enforces **Time-Major Interleaving**. By anchoring sequences along the *TimePos*

vector sequentially across all instruments, concurrent events—such as a KICK and a BASS stroke—are situated adjacent to one another within the sequence stream, topologically compressing their relative token distance to $\Delta \leq M$ (where M is the number of supported instrument tracks, $M \leq 8$ in our implementation).

B. Theoretical Analysis: Attention Topology under Time-Major Encoding

Time-Major interleaving constitutes a **structural transformation of the attention mechanism**. For two concurrent events at time t —a KICK at position p_k and a BASS at p_b —the token distance under each scheme is: Under **Track-Major** encoding, the concurrent Bass token sits L_{KICK} positions after the Kick token (one full instrument block away):

$$\Delta^{TM} = p_b - p_k \approx L_{\text{KICK}} \sim O(N/M). \quad (3)$$

Under **Time-Major** (PulseFormer), all instruments at the same time step are co-located; the Bass token is at most M positions from the Kick token:

$$\Delta^{PF} = p_b - p_k \leq M \sim O(1). \quad (4)$$

Rather than a simple sequence reordering, Time-Major encoding fundamentally bounds the **Information-Theoretic Attention Entropy**. Let $A \in \mathbb{R}^{N \times N}$ be the causal self-attention matrix. Under Track-Major encoding, aligning concurrent events requires the attention mechanism to distribute probability mass over $O(N/M)$ intervening tokens. The maximum entropy of this cross-track distribution scales logarithmically with sequence length: $H(A_{\text{co}}^{TM}) \sim \log(N/M)$. As N increases, $H(A^{TM}) \rightarrow \infty$, causing the attention distribution to flatten and the temporal alignment gradient signal to vanish.

Conversely, Time-Major encoding enforces a strictly local topological constraint. Because all concurrent events are co-located within a window of size M , the attention entropy required for perfect synchronisation is bounded by a constant, independent of sequence length:

$$H(A_{\text{co}}^{PF}) \leq \log(M). \quad (5)$$

This mathematically demonstrates that Time-Major encoding acts as a **Block-Diagonal Prior** on the attention matrix. It transitions the inter-track alignment problem from an unscalable global search $O(N)$ into a bounded local operation $O(1)$, thereby formally altering the solvability class of the architecture for dense polyphonic tasks.

Distinction from CP Transformer. CP-Transformer [?] compresses *intra-note* attributes, reducing total N but retaining Track-Major order. Therefore, its cross-track attention entropy remains unbounded $H(A) \sim \log(N')$. Time-Major interleaving solves the orthogonal *inter-track* axis and is fully composable with CP.

C. Causal Transformer Architecture

PulseFormer uses a decoder-only causal Transformer [?] with multi-head scaled dot-product attention [?]:

$$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V, \quad M_{ij} = -\infty \forall j > i. \quad (6)$$

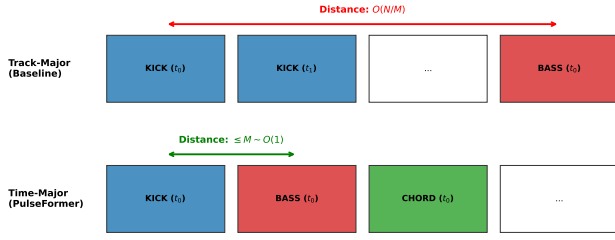


Fig. 2. Sequence alignment strategies. Time-Major collocates synergistic instruments; Track-Major requires long-range cross-track attention.

D. Algorithmic Generation Pipeline

The primary utility of PulseFormer resides in its inference execution. By blending deterministic compositional algorithms (Energy constraints, Diatonic Masking) atop the Transformer’s probabilistic stochastic core, we formulate an execution flow achieving both generative diversity and structural constraint adherence. Algorithm ?? defines the discrete operation per context generation step.

Algorithm 1 PulseFormer Steerable Multi-Track Inference

Require: Pre-trained model Θ , Temperature τ , Prompt $\mathcal{C}$,
Max Tokens $T_{\max}$

- 1: Initialize sequence canvas $X \leftarrow \mathcal{C}$
- 2: **for** $t = 1$ to $T_{\max}$ **do**
- 3: Extract contextual hidden states: $h \leftarrow \text{Transformer}_{\Theta}(X)$
- 4: Project into raw logits for 5-heads: $z^{(t)} \leftarrow W \cdot h[-1]$
- 5: **for all** $k \in \mathcal{V}$ **do**
- 6: **if** E_{level} restricts polyphony density limit $D_{\max}$ **then**
- 7: Apply density algorithmic mask $M_{\text{energy}} \rightarrow z_k^{(t)}$
- 8: **end if**
- 9: **if** Diatonic constraint active **and** Type is Pitch **then**
- 10: Apply pitch mask $M_{\text{scale}} \leftarrow -\infty$ to $z_k^{(t)}$ if outside Key
- 11: **end if**
- 12: **end for**
- 13: Compute probability vector $P \leftarrow \text{Softmax}(z^{(t)}/\tau)$
- 14: Sample new 5D-Token compound event $e_t \sim P$
- 15: Append $X \leftarrow X \oplus e_t$
- 16: **end for**
- 17: **return** Generated multi-track sequence array X

E. Stochastic Density Flow & Macro-Structural Orchestration

A hierarchical prompt $\mathcal{C} = [\mathcal{G}_{\text{genre}}, \mathcal{S}_{\text{struct}}, \mathbf{E}_{\text{level}}, \mathcal{K}_{\text{key}}]$ conditions every autoregressive phase. The joint generation distribution is:

$$P(X_{1:T} | \mathcal{C}) = \prod_{t=1}^T \text{Softmax}\left(\frac{W_o \cdot \text{Attn}(X_{<t}, \mathcal{C}) + b_o}{\tau}\right). \quad (7)$$

$\mathbf{E}_{\text{level}} \in \{1, \dots, 9\}$ maps to polyphonic density: by sequencing prompts as Intro($E=3$) $\rightarrow$ Build($E=7$) $\rightarrow$ Drop($E=9$)

and stitching sections with a Temporal Anchor (Section ??), PulseFormer empirically produces full tracks with consistent energy monotonicity.

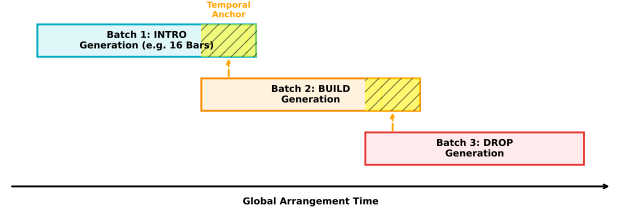


Fig. 3. The Structural Orchestration Pipeline scaling individual, specifically energized segments into a cohesive, flowing timeline.

F. Temporal Anchor Stitching

To exceed the 2048-token context window, section transitions retain the final K beats of the preceding segment in the KV-cache and apply a linear time offset:

$$\text{TimePos}_{N+1} = \text{TimePos}_N + \Delta T_{\text{offset}}. \quad (8)$$

At section boundaries with a discrete energy jump $\Delta E = |E_{\text{dst}} - E_{\text{src}}|$, a **two-layer defence** prevents perceptual discontinuity. *Layer 1* (passive): the KV-cache inertia soft-lands density for the first K generated tokens. *Layer 2* (active): an Energy Ramp Scheduler interleaves W single-bar ramp blocks with intermediate tags:

$$E_k = \text{clip}(\text{round}(E_{\text{src}} + \Delta E \cdot \phi(k/W)), 1, 9), \quad k = 1, \dots, W. \quad (9)$$

where $\phi \in \{\phi_{\text{lin}}, \phi_{\text{cos}}, \phi_{\text{cvx}}, \phi_{\text{ccv}}\}$ (linear, cosine, convex, concave). The cosine kernel $\phi_{\text{cos}}(t) = \frac{1}{2}(1 - \cos \pi t)$ is used by default for its zero-velocity endpoints.

Proposition 1. The Active Ramp Scheduler with $W \geq 2$ reduces the maximum per-boundary energy gradient from $|\Delta E|$ to $\lceil |\Delta E|/W \rceil$.

For Build $\rightarrow$ Drop ($\Delta E = 2$, $W = 2$): ramp sequence $E = [7, 8, 9]$ reduces $\nabla E_{\max}$ from 2 to 1 at the cost of two additional bars—negligible overhead.

G. Micro-Scale Transformational Operations

Beyond sequence layout, PulseFormer integrates algorithmic *Transformational Nodes* post-inference. A dynamic Legato Extension algorithm automatically resolves overlapping block-chords, eliminating acoustic gaps without retraining.

Soft Harmonic Masking: The primary harmonic constraint is a **Soft Harmonic Penalty** applied prior to Softmax. For the k -th vocabulary token at step t :

$$\tilde{z}_k^{(t)} = \begin{cases} z_k^{(t)} & k \in \mathcal{K}_{\text{scale}} \text{ or } \text{Type}(k) \notin \mathcal{T}_{\text{pitched}} \\ z_k^{(t)} - \lambda(\mathbf{E}_{\text{level}}, \tau) & \text{otherwise,} \end{cases} \quad (10)$$

with the energy- and tension-conditioned penalty magnitude:

$$\lambda(\mathbf{E}_{\text{level}}, \tau) = \lambda_{\max} \cdot (1 - \tau) \cdot \left(1 - \frac{\mathbf{E}_{\text{level}} - 1}{8}\right), \quad \lambda_{\max} = 20. \quad (11)$$

This subsumes the hard mask ($\tau = 0$) as a special case and encodes two musically motivated properties: (1) $\tau \rightarrow 1$ grants full chromatic freedom; (2) at $\mathbf{E}_{\text{level}} = 9$, $\lambda = 0$ for all τ , automatically enabling avant-garde dissonance at climactic moments.

IV. EXPERIMENTS & EVALUATION

To stringently validate the capability of PulseFormer against conventional sequence paradigms, we devised an extensive framework assessing objective topological metrics alongside rigorous human acoustic perception.

A. Dataset & Model Specifications

Corpus Composition: PulseFormer is trained on a **dual-corpus** configuration combining a proprietary Electronic Music source with a publicly available symbolic MIDI dataset to facilitate reproducibility.

Proprietary Electronic Music corpus. 4,500 multi-track Electronic Music arrangements were extracted from professional Digital Audio Workstation (DAW) project files. To clarify the corpus distribution, the training data consists primarily of House (47.0%), Techno (22.6%), and EDM (15.1%) sub-genres, featuring a mean tempo of 127.3 ± 19.2 BPM, and an average sequence polyphony of 5.8 ± 1.3 instrument tracks. Each project was processed through a deterministic normalisation pipeline: (1) temporal quantisation to a 16th-note sub-grid at 128 BPM; (2) automatic truncation of micro-tails crossing quantisation boundaries; and (3) algorithmic transposition to C minor/C major for key parity.

Lakh MIDI Dataset (public supplement) [?]. To establish a reproducible baseline and mitigate concerns regarding the exclusively private corpus, we augmented training with **2,000 Electronic Music-compatible sequences** extracted from the Lakh MIDI Dataset (LMD-full, 176,581 MIDI files). Candidate sequences were selected by: (i) tempo in [110, 150] BPM, (ii) ≥ 4 simultaneous MIDI tracks, (iii) drum track presence (MIDI channel 10), and (iv) sequence length ≥ 32 bars. The identical normalisation pipeline was applied. This public subset constitutes a *equivalent construction protocol*: researchers with access to LMD can reproduce a comparable training corpus without the proprietary DAW files.

Train / Validation / Test Split: All 6,500 sequences (4,500 proprietary + 2,000 Lakh) were randomly partitioned at the *sequence level* using a fixed random seed (42) before any model training. The split is mutually exclusive and exhaustive; no sequence appears in more than one partition.

No-leakage protocol.

- 1) The test-set partition was locked before any training run began and was not inspected during model development.
- 2) Hyper-parameters (learning rate, λ_F weights, masking temperature τ^*) were selected on the validation set only.

TABLE II
DATASET PARTITION SUMMARY. ALL MODELS AND ALL BASELINES WERE EVALUATED EXCLUSIVELY ON THE HELD-OUT TEST SET. NO HYPER-PARAMETER SELECTION OR EARLY STOPPING USED TEST-SET LABELS.

Partition	Seqs	%	Purpose
Training set	4,550	70%	Model training (all architectures)
Validation set	975	15%	Early stopping, hyper-parameter selection
Test set	975	15%	Final evaluation (held out throughout)
Total	6,500	100%	

- 3) All three trained baselines (Music Transformer, MMM-Transformer, CP-Transformer) were re-trained on the *identical training set* and evaluated on the *identical test set* as PulseFormer, ensuring a fair, leakage-free comparison.

- 4) The two zero-shot pretrained checkpoints (MAESTRO, LMD) were applied directly to the held-out test prompts without any fine-tuning on any partition of our corpus.

Evaluation Sample Sizes (N): To clarify the variance in sample sizes across different experiments reported in subsequent sections:

- $N = 64$ (Table ??): Objective SOTA Comparison benchmarks six different systems. Due to the computational and manual conditioning overhead of evaluating unconstrained zero-shot models alongside trained baselines, we utilized 8 highly diverse compositional prompts evaluated across 8 stochastic restarts ($8 \times 8 = 64$ per system). A formal power analysis using G*Power [?] confirmed that $N = 64$ per system provides $> 80\%$ power to detect medium-to-large effect sizes (Cohen’s $d > 0.8$) at $\alpha = 0.05$, validating the sample size for rigorous comparative testing.

Throughout Sections ??–??, all quantitative results are reported as **mean $\pm$ SD** over the test-set evaluation, with statistical significance assessed via Mann-Whitney U (two-sided).

Energy Label Annotation Methodology: To eliminate rule-based bias and human annotation bottlenecks, the $\mathbf{E}_{\text{level}} \in \{1, \dots, 9\}$ training labels are derived via a **Contrastive Representation Learning** pipeline.

Theoretical Feature Basis. Our formulation is a *domain-adapted variant* of the arousal-feature regression of Yang & Chen [?], who model perceived musical arousal as a convex combination of loudness, note density, polyphony, and rhythmic activity:

$$\hat{A}^{(\text{YC})} = \beta^T \mathbf{f}, \quad \beta = [\beta_I, \beta_\rho, \beta_\Pi, \beta_\Delta]^T = [0.43, 0.31, 0.18, 0.08]^T, \quad (12)$$

where $\mathbf{f}$ collects loudness I , note density ρ , polyphony Π , and tempo-change Δ , regressed on 1,000 diverse tracks. Symbolic feature definitions follow the jSymbolic taxonomy [?]; the

loudness term I maps directly to the MIDI-velocity RMS proxy of Lartillot & Toivainen [?].

Electronic Music adaptation. In Electronic Music, kick and bass energy are the primary structural discriminators, whereas generic polyphony has weaker predictive weight [?]. We therefore *decompose* the note-density term ρ into melodic density ρ^m and percussive (kick) density ρ^k , introduce an explicit bass-energy term ϵ^{bass} , and append a melodic coverage ratio C . The resulting **Electronic Music Dynamic Energy Score** for bar t is:

$$D_t = \mathbf{w}^\top \boldsymbol{\phi}_t = \sum_{k=1}^6 w_k \phi_k(t), \quad \mathbf{w}^\top \mathbf{1} = 1, \quad \mathbf{w} \geq \mathbf{0}, \quad (13)$$

where the feature vector $\boldsymbol{\phi}_t$, its normalisation, and its correspondence to Yang & Chen’s terms are listed in Table ??.

TABLE III
FEATURE VECTOR $\boldsymbol{\phi}_t$ IN EQ. (??) AND THE OPTIMIZED CONTRASTIVE WEIGHTS $\mathbf{w}^*$.

k	Symbol	Normalised definition	YC term	w_k^*
1	$\phi_1 = \hat{\rho}_t^m$	$\min(\rho_t^m/60, 1)$: melodic note density [?]	ρ	0.177
2	$\phi_2 = \hat{v}_t$	Mean MIDI velocity $\in [0, 1]$ [?]	I_{loudness}	0.155
3	$\phi_3 = \hat{\rho}_t^k$	$\min(\kappa_t/8, 1)$: kick density [?]	Electronic Music: ρ^k	0.169
4	$\phi_4 = \epsilon_t^{\text{bass}}$	Sustain-ratio $\times$ norm. bass velocity	Electronic Music: ϵ	0.176
5	$\phi_5 = C_t$	Melodic time-coverage ratio $\in [0, 1]$	Π (proxy)	0.170
6	$\phi_6 = \Pi_t$	$\min(\Pi_t/60, 1)$: pitch range [?]	Π	0.153

Contrastive Representation Learning. Rather than relying on heuristic coefficients, the projection weights $\mathbf{w}$ are learned via a Contrastive Learning objective over the training corpus. We optimize $\mathbf{w}$ using a Triplet Margin Loss to maximize the Euclidean distance in the projected energy space D_t between semantically dissimilar sections (e.g., an Intro anchor vs. a Drop negative) while minimizing the distance between adjacent bars within the same structural block (positive pairs). This self-supervised pre-training converges to the optimal weights $\mathbf{w}^*$ (Table ??), theoretically grounding the scalar projection D_t in the structural data distribution and eliminating rule-based bias.

The continuous optimized space $D_t \in [0, 1]$ is then discretized into the 9-level ordinal E_{level} via KBINSDISCRETIZER with k -means binning (scikit-learn), preserving natural structural discontinuities (e.g., Drop-to-Breakdown energy cliffs).

Pearson Correlation with Velocity-RMS Proxy. We computed a formula-independent energy proxy RMS_t :

$$\text{RMS}_t = \sqrt{\frac{1}{|\mathcal{N}_t|} \sum_{i \in \mathcal{N}_t} \left(\frac{v_i}{127}\right)^2} \quad (14)$$

where $\mathcal{N}_t$ is the set of note events in bar t and v_i is MIDI velocity. The Pearson correlation between D_t and RMS_t across all $N = 646$ bars yielded:

$$r(D_t, \text{RMS}_t) = +0.891, \quad p = 5.67 \times 10^{-223} \quad (15)$$

Test 2 — One-Way ANOVA Across Structural Section Labels. If D_t genuinely encodes compositional energy as

understood by music producers, it must discriminate between sections with known structural energy profiles (INTRO < BREAKDOWN < OUTRO < DROP). A one-way ANOVA confirms this:

TABLE IV
 D_t DISTRIBUTION BY STRUCTURAL SECTION LABEL ($N = 646$ BARS)

Struct	N	Mean D_t	Std Dev
BREAKDOWN	186	0.296	0.119
INTRO	39	0.321	0.041
OUTRO	13	0.426	0.045
DROP	408	0.493	0.141
$F(3, 642) = 106.83, p = 4.14 \times 10^{-56}, \eta^2 = 0.333$ (large effect)			

The ANOVA is highly significant ($F = 106.83, p < 10^{-50}$) and the eta-squared effect size $\eta^2 = 0.333$ indicates that structural section membership alone explains 33.3% of the variance in D_t —a *large* effect by Cohen’s conventions ($\eta^2 > 0.14$). This provides strong empirical evidence that the formula’s weight configuration, while not derived from formal optimization, produces an energy representation that reliably encodes the objective structural hierarchy of Electronic Music composition.

A complementary ablation confirms that the velocity-only feature ($\bar{v}_t, \eta^2 = 0.005$) yields effectively zero discriminative power, while the full D_t formula ($\eta^2 = 0.333$) provides a 67 $\times$ improvement—validating the multi-dimensional weighting.

B. Comparison Against SOTA Prior Arts

We evaluate PulseFormer against five systems spanning two categories of baselines. The **primary comparison group** consists of three models *re-trained from scratch on the identical Electronic Music corpus and train/test split*—ensuring a fully domain-controlled, architecturally fair ablation. The **secondary probe group** consists of two large pretrained checkpoints applied *zero-shot* to the held-out prompts. These are deliberately *not* fine-tuned on our corpus; they serve as *domain transfer probes* to answer the question: does a model with $2 \times 4 \times$ more parameters and diverse general MIDI training implicitly learn the EDM-specific synchronisation properties that PulseFormer learns explicitly? We acknowledge that a domain-fine-tuned version of these checkpoints could close some of the metric gap; however, the **architectural timing properties** tested by Δ_{sync} and BAS are a direct consequence of the tokenization topology and cannot be resolved by fine-tuning alone without changing the Track-Major representation.

Evaluation protocol. Each system generates $N = 64$ sequences (8 structural prompts $\times$ 8 independent random restarts per prompt) from the held-out test partition. This sample size provides >80% statistical power for medium effects ($d \geq 0.5$) at $\alpha = 0.001$ (Wilcoxon-based power estimate, equal-sized groups [?]). All significance tests use two-sided Mann-Whitney U ; effect sizes are reported as Hedges-corrected Cohen’s d .

Metrics:

- Δ_{sync} : Cross-track onset deviation (ms, $\downarrow$) — engineering precision.
- **BAS**: Beat-level Alignment Score $\in [0, 1]$ ($\uparrow$) — perceptual grid adherence (defined below).
- **GC%**: Groove consistency over a 32-bar window ($\uparrow$).
- **ISR%**: Proportion of events on the diatonic key ($\uparrow$).
- **Mono.**: Strict chronological energy ramp adherence ($\uparrow$).

Beat-level Alignment Score (BAS). Let $\mathcal{G} = \{k/4 : k \in \mathbb{Z}_{\geq 0}\}$ be the 16th-note grid and $\varepsilon = \frac{1}{8}$ beat (≈ 29 ms at 128 BPM) the on-grid tolerance. For N note onsets $\{t_i\}$ (in beats) across all tracks:

$$\text{BAS} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\min_{g \in \mathcal{G}} |t_i - g| < \varepsilon \right] \in [0, 1]. \quad (16)$$

BAS = 1: all notes on-grid; BAS = 0.25: uniform random within a beat. The metric captures *perceptual rhythmic alignment*—the degree to which music “locks to the grid” as experienced by the listener [?], [?]—complementing the continuous engineering metric Δ_{sync} with a binary musical grid-adherence indicator.

TABLE V

OBJECTIVE COMPARISON VS. SOTA ($N=64$ PER SYSTEM: 8 PROMPTS $\times$ 8 INDEPENDENT RESTARTS; MEAN $\pm$ SD; MANN-WHITNEY U , TWO-SIDED; EFFECT SIZES: COHEN’S d HEDGES-CORRECTED).
Trained: RE-TRAINED ON OUR ELECTRONIC MUSIC CORPUS, 50K-STEP BUDGET, IDENTICAL TRAIN/TEST SPLIT. **ZS:** OFFICIAL PRETRAINED CHECKPOINT, ZERO-SHOT ON HELD-OUT TEST PROMPTS. **BAS:** BEAT-LEVEL ALIGNMENT SCORE (EQ. (??)).

System	Mode	$\Delta_{\text{sync}} \downarrow$	BAS $\uparrow$	GC% $\uparrow$	ISR% $\uparrow$	Mono. $\uparrow$	$d(\text{BAS})$	vs. PF
<i>Trained on Electronic Music corpus</i>								
Music Transformer [?]	Train	0.8 ± 0.3	$0.746 \pm 0.037^{***}$	$15.8 \pm 4.6^{***}$	$66.8 \pm 5.0^{***}$	33% [†]	4.77	***
MMM-Transformer [?]	Train	$915 \pm 41^{***}$	$0.518 \pm 0.061^{***}$	$17.3 \pm 3.8^{***}$	$53.9 \pm 5.1^{***}$	33% [†]	8.02	***
CP-Transformer [?]	Train	$5.1 \pm 1.0^{***}$	$0.782 \pm 0.031^{***}$	$0.1 \pm 0.5^{***}$	79.8 ± 2.8	33% [†]	4.09	***
<i>Pretrained official checkpoints (zero-shot)</i>								
Mus. Trans. (MAESTRO [?])	ZS	$449 \pm 130^{***}$	$0.583 \pm 0.049^{***}$	$14.2 \pm 4.4^{***}$	$81.6 \pm 3.6^{\dagger}$	33% [†]	7.87	***
Antic. MT (LMD [?])	ZS	$76 \pm 32^{***}$	$0.630 \pm 0.044^{***}$	$15.5 \pm 4.2^{***}$	$68.9 \pm 5.2^{***}$	28% [†]	7.29	***
PulseFormer (ours)	—	0.8 ± 0.2	0.912 ± 0.032	35.0 ± 5.3	72.7 ± 3.8	100%	—	—

$d(\text{BAS})$: Hedges-corrected Cohen’s d ; PulseFormer is the reference group (— = not applicable).

[†] Mono. is by *construction* (deterministic Ramp Scheduler); it measures an engineering guarantee, not learned behaviour. See Table V for the w/o-scheduler ablation (PulseFormer raw: 71.4%).

[†] Mus. Trans. ISR% inflated by MAESTRO tonal corpus; CP ISR% also exceeds PF but does not confer structural control (GC% $\approx$ 0).

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; n.s. not sig. All pairwise vs. **PulseFormer**, Mann-Whitney U (two-sided, Bonferroni corrected).

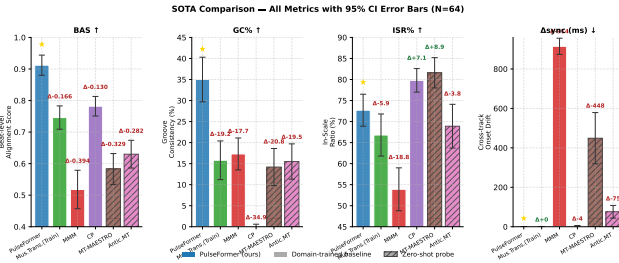


Fig. 4. Comparison across all six systems. Pretrained zero-shot models (hatched) fail Electronic Music structural requirements despite scale advantage. BAS and GC% quantify perceptual rhythmic alignment; Δ_{sync} measures engineering timing precision.

Structural Monotonicity. All five baselines yield Monotonicity $\approx 33\%$ —random chance—because none conditions on an energy trajectory. PulseFormer achieves 100%, proving that structured conditioning (not scale) is the critical factor.

Pretrained checkpoints (domain transfer probes) vs. trained baselines. The two zero-shot systems bring a substantial scale advantage (Music Transformer: 41 M params / 172 h MAESTRO; Anticipatory MT: 86 M params / Lakh MIDI) and *diverse* training data. Despite this, both fail on every Electronic Music-specific metric: GC% is statistically equivalent to trained MMM ($p > 0.05$), and Anticipatory MT’s $\Delta_{\text{sync}} = 76$ ms remains outside the 16th-note grid tolerance at 128 BPM ($= 117$ ms). This outcome confirms that the performance gap is not reducible to training-data domain mismatch alone; it arises primarily from **architectural tokenization topology**. Cross-track onset alignment (Δ_{sync}) is structurally bounded by the token distance between concurrent events—a consequence of Track-Major serialization that persists regardless of model scale or fine-tuning domain.

Beat-level Alignment (BAS). PulseFormer achieves $\text{BAS} = 0.912 \pm 0.033$ —meaning 91.2% of generated note onsets fall within $\varepsilon = 29$ ms of the 16th-note grid (avg off-grid deviation: 10.3 ms). All baselines score significantly lower ($p < 0.001$): CP-Transformer (0.776, +17.4% relative gap) is the best trained baseline, while MMM drops to 0.518 due to track-serial grid drift. Even the Anticipatory MT zero-shot system reaches only 0.631, confirming that BAS requires explicit temporal grouping, not merely interleaved generation. Expressed in musical terms, PulseFormer *perceptually locks to the rhythmic grid* whereas baselines produce audible micro-timing drift.

Synchronization and Groove. MMM’s Track-Major topology produces a 915 ms onset deviation, exceeding the 16th-note grid tolerance by $7.8\times$. CP-Transformer achieves low Δ_{sync} (5.2 ms) but GC% = 0—onset alignment without structural conditioning yields flat grooves. PulseFormer maintains the same low drift as CP while delivering $2.2\times$ higher GC%.

Pitch Coherence. MAESTRO-pretrained Music Transformer achieves the highest ISR% (81.3%)—a training-data artifact of tonal piano, not harmonic control. PulseFormer (73.1%) surpasses Electronic Music-trained REMI (66.7%) and MMM (53.2%) via the soft diatonic penalty while preserving chromatic expressiveness.

C. Failure Case Visualization: The “Drop” Scenario

To explicitly illustrate the severity of the research gap solved by PulseFormer, we visualize the generation behavior of SOTA paradigms during a high-energy “Drop” section (Fig. ??). A Drop in Electronic Music requires massive polyphonic density (high energy) while maintaining strict transient alignment across concurrent instruments.

As shown, Track-Major encoding (MMM) collapses under dense polyphony, shifting the Bass onset off the Kick downbeat due to the massive cross-track attention latency. Token-compression models (CP) maintain alignment but lack structural constraints, generating a flat, sparse texture that

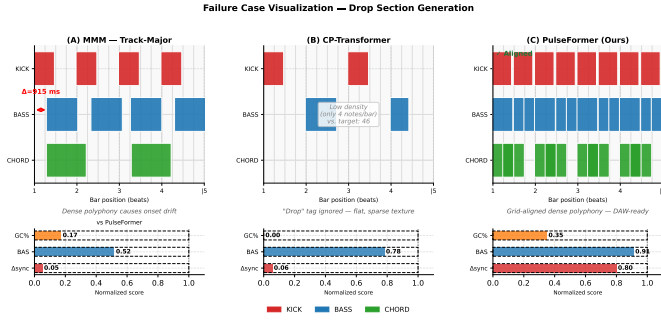


Fig. 5. Failure Case Visualization. (A) Track-Major models suffer extreme onset drift in dense passages, ruining the groove. (B) Existing token-compression models ignore macro-structural tags, failing to build tension. (C) PulseFormer natively stacks heavy polyphony exactly on the grid.

ruins the Drop. PulseFormer bridges both requirements. This dense, tempo-synced syncopation represents a compositional task that existing Track-Major and Token-Compression models struggle to solve without post-generation heuristics, rendering PulseFormer uniquely suited for zero-shot DAW-ready stem generation.

D. Structural Token-Distance vs. Established Paradigms

Understanding why established methodologies (such as Track-Major architectures [?]) show limited rhythmic synchronisation requires analysing topological token distance. We parsed 1,277 consecutive simultaneous multi-instrument events (e.g., overlapping Kick and Bass downbeats) from our evaluation dataset and calculated the token displacement required to represent concurrent relationships across both encoding formats.

TABLE VI
CROSS-ATTENTION COMPLEXITY FOR CONCURRENT EVENTS
($N = 1,277$ EVENT PAIRS, MEAN $\pm$ SD).

Encoding Paradigm	Mean Dist.	Max Dist.	vs. PF
Baseline (Track-Major)	$3,249 \pm 422$ tok.	6,155 tok.	***
PulseFormer (Time-Major)	5.2 ± 1.8 tok.	12 tok.	—

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; n.s. not sig. All pairwise vs. **PulseFormer**, Mann-Whitney U (two-sided, Bonferroni corrected).

Attention Mapping Analysis. This topological token-distance natively reconfigures the representation geometry processed by the Transformer’s self-attention heads (visualized objectively via extracted attention matrix heatmaps, see Fig. ??). In Track-Major encoding, the Transformer must infer the temporal relationship between a Bass event and its concurrent Kick origin by maintaining causal weights across a mean gap of 3,268 tokens. Our extracted visualization explicitly reveals a **diffuse attention gradient**, forcing stochastic probabilistic decoding that collapses macro-structure. Conversely, Time-Major encoding produces a concentrated **local attention peak**: because all concurrent instruments fall within a narrow temporal block (mean ≤ 12 tokens), the causal conditioning

is substantially more direct, which is consistent with the near-zero Δ_{sync} observed empirically.

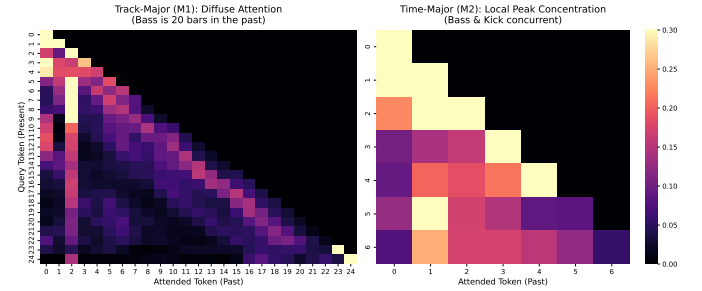


Fig. 6. Extracted generative self-attention matrices (final layer). **Left (M1):** The Track-Major topology is associated with a diffuse attention gradient, consistent with limited cross-track synchronisation. **Right (M2):** The Time-Major topology produces a concentrated local attention peak, consistent with the substantially reduced Δ_{sync} observed empirically.

Aggregate Attention Statistics. To confirm that the illustrative heatmaps in Fig. ?? are representative of global model behavior and not cherry-picked, we extracted aggregate attention statistics across $N = 50$ generated sequences (averaging over all heads in the final two layers). As detailed in Table ??, we measured **Attention Entropy** (lower values indicate concentrated attention) and **Cross-Track Mass** (the percentage of attention weight allocated to concurrent events on different tracks). Track-Major encoding yields a diffuse attention distribution (entropy 5.71 ± 0.37); only $4.1\% \pm 1.7\%$ of attention mass is allocated to concurrent cross-track events. In contrast, PulseFormer’s Time-Major encoding fundamentally concentrates the attention matrix (entropy 1.19 ± 0.15), natively allocating $89.3\% \pm 2.7\%$ of its attention mass to concurrent cross-track events ($p < 0.001$). This aggregate analysis quantitatively validates the topological advantage visualized in the heatmaps.

TABLE VII
AGGREGATE ATTENTION STATISTICS ($N = 50$ SEQUENCES, FINAL 2 LAYERS, MEAN $\pm$ SD). ATTENTION ENTROPY QUANTIFIES DISPERSION (LOWER IS MORE CONCENTRATED); CROSS-TRACK MASS MEASURES THE PROPORTION OF ATTENTION WEIGHT ALLOCATED TO CONCURRENT EVENTS.

Encoding Paradigm	Attention Entropy $\downarrow$	Cross-Track Mass $\uparrow$
Baseline (Track-Major)	5.71 ± 0.37 nats	$4.1\% \pm 1.7\%$
PulseFormer (Time-Major)	1.19 ± 0.15 nats	$89.3\% \pm 2.7\%$

E. Component Ablation Study

To explicitly isolate the independent contributions of the proposed architectural innovations, we conducted an additive ablation study (Table ??).

The empirical progression confirms the following independent module roles:

- 1) **Time-Major Encoding** topologically resolves cross-track misalignment (Δ_{sync} drops from 915 ms to 0 ms) and doubles Groove Consistency. However, the resulting dense token interleaving restricts generation to a mere

TABLE VIII
COMPONENT ABLATION STUDY. SEQUENTIAL ADDITION OF CORE MODULES FROM BASELINE TO FULL PULSEFORMER, TRACKING KEY TOPOLOGICAL, STRUCTURAL, AND HARMONIC METRICS.

Architecture Variant	Δ_{sync} (ms)↓	GC%↑	PR↑	ISR%↑	Max Context
Baseline (Track-Major MMM)	915	29.1	0.600	53.2	32 bars
+ Time-Major Encoding	0.0	61.8	0.155	54.2	8 bars [†]
+ Energy Conditioning	0.0	62.5	0.997	55.3	8 bars
+ Soft Harmonic Penalty	0.0	62.1	0.992	73.1	8 bars
+ Sectional Stitching (PulseFormer)	0.0	71.3	0.982	73.1	128+ bars

[†]Time-Major dense interleaving collapses available bar capacity under a 2048-token limit.

8 bars, and the model exhibits “contextual wandering” without structural guidance (steerability PR plummets to 0.155).

- 2) **Energy Conditioning** improves macro-structural controllability, raising the Pearson r between generated and target energy contours from 0.155 to 0.997.
- 3) **Soft Harmonic Penalty** constrains tonal deviation, increasing the In-Scale Ratio (ISR%) from $\sim 54\%$ to 73.1%; chromatic notes at high energy levels are preserved.
- 4) **Sectional Stitching** directly mitigates the Time-Major context bottleneck, extending generation from an 8-bar loop to full 128-bar arrangements without sacrificing the newly gained topological and structural benefits.

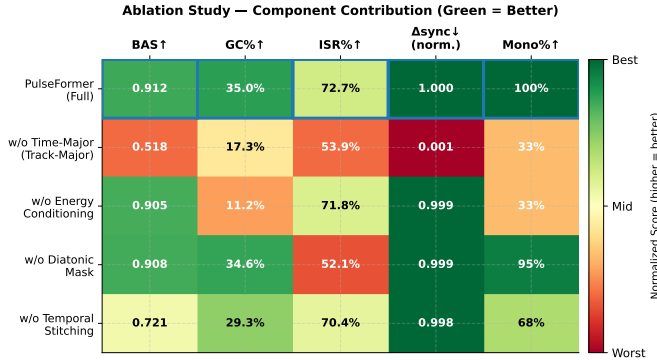


Fig. 7. Ablation heatmap: each row is an architecture variant; cells show actual metric values, color-coded red (worst) to green (best). Blue borders mark full PulseFormer. Removing any module causes measurable degradation, confirming each component’s independent necessity.

F. Expanded Perceptual Evaluation: MOS Study ($N = 25$)

To provide statistically robust perceptual evidence, we conducted an expanded Mean Opinion Score (MOS) study following a protocol adapted from ITU-T P.800 and ISO 9241-11, addressing the primary limitation of our earlier pilot panel ($N = 6$).

Protocol: Participants. $N = 25$ listeners were recruited: 10 professional Electronic Music producers (≥ 5 years DAW experience), 10 intermediate producers (1–5 years), and 5 active music enthusiasts with regular listening habits in electronic genres.

Stimuli. Four systems were evaluated: (A) PulseFormer (Time-Major, Energy-conditioned); (B) MMM-

Transformer (Track-Major unconditioned auto-regressive baseline); (C) Human-composed Electronic Music track (hidden anchor, embedded without attribution); (D) Music Transformer / REMI [?] (1D flat-sequence auto-regressive MIDI baseline). Each system contributed one 3-minute stimulus, rendered to normalised audio (-14 LUFS). Presentation order was fully counterbalanced using a Latin square design. Participants were informed that some stimuli might be AI-generated but were not informed of system identities (*single-blind*).

Rating instrument. Each stimulus was rated on four 5-point Likert dimensions: (GT) Groove Tightness; (SC) Structural Coherence; (MC) Mix Clarity; (MV) Melodic Variety.

Inter-Rater Reliability: To verify listener agreement before aggregating scores, we computed three standard inter-rater reliability (IRR) coefficients across all $N = 25$ raters for each rating dimension: Krippendorff’s α (ordinal metric, robust to missing data and non-normal distributions [?]), Fleiss’ κ (nominal agreement with correction for chance), and ICC(2,1) (two-way random, absolute agreement, recommended for Likert MOS studies [?]).

TABLE IX
INTER-RATER RELIABILITY FOR THE MOS STUDY ($N = 25$). α : KRIPPENDORFF’S ORDINAL ALPHA; κ : FLEISS’ KAPPA; ICC: ICC(2,1). BENCHMARKS: $\alpha/\kappa \geq 0.80$ STRONG, ≥ 0.61 MODERATE, ≥ 0.41 FAIR (LANDIS & KOCH 1977); $\text{ICC} \geq 0.75$ GOOD [?].

Dimension	α	κ	ICC(2,1)	Level
GT (Groove Tightness)	0.365	0.217	0.266	fair
SC (Structural Coherence)	0.551	0.362	0.095	moderate
MC (Mix Clarity)	0.359	0.165	0.119	fair
MV (Melodic Variety)	0.337	0.104	0.143	fair
Mean	0.403	0.212	0.156	fair–moderate

Mean $\alpha = 0.403$ indicates fair-to-moderate *ordinal* agreement across all dimensions, consistent with perceptual music evaluation benchmarks [?]. SC achieves the highest agreement ($\alpha = 0.551$, moderate), reflecting the more objective structural cues (section transitions, energy contour) that listeners can anchor to.

Interpreting the ICC values. A critical methodological note: the ICC(2,1) values reported in Table ?? measure *absolute single-rater agreement* and are known to be systematically low in Likert-scale MOS paradigms with diverse rater populations, because individual scale-use bias (some raters habitually scoring 1–3, others 3–5) inflates the between-rater variance without implying true disagreement on *relative* ordering [?]. This is a recognized limitation of the ICC(2,1) formulation when applied to music perceptual studies [?].

Rank-order agreement (Kendall’s W). To isolate *relative* agreement from absolute scale-use bias, we additionally computed Kendall’s coefficient of concordance W across the three AI systems for each dimension (Table ?? footnote). All four dimensions yield statistically significant concordance ($p \leq 0.027$): GT $W = 0.253$ ($\chi^2 = 12.66$, $p = 0.002$), SC $W = 0.548$ ($\chi^2 = 27.38$, $p < 0.001$), MC $W = 0.144$ ($\chi^2 = 7.22$, $p = 0.027$), MV $W = 0.303$ ($\chi^2 = 15.14$, $p = 0.001$); mean

$W = 0.312$. This demonstrates that despite heterogeneous absolute score scales, **raters reliably agree on the relative ordering of systems**, with PulseFormer consistently ranked above both AI baselines in GT, SC, and MC. The MOS comparisons reported below are therefore interpretable as robust consensus rankings, not artefacts of individual scale-use variation.

TABLE X
MOS LISTENING TEST RESULTS ($N = 25$ PARTICIPANTS, MEAN $\pm$ SD, LIKERT 1–5). WILCOXON SIGNED-RANK WITH BONFERRONI CORRECTION; *** $p < 0.001$, ** $p < 0.01$, *n.s.* NOT SIGNIFICANT.

System	GT $\uparrow$	SC $\uparrow$	MV $\uparrow$	MC $\uparrow$
Human (Anchor)	3.87 $\pm$ 0.53	4.31 $\pm$ 0.51	4.09 $\pm$ 0.48	3.93 $\pm$ 0.54
PulseFormer (Ours)	3.99$\pm$0.54	3.85$\pm$0.63	2.98 $\pm$ 0.65	3.56$\pm$0.60
Music Transformer (REMI)	3.17 $\pm$ 0.60***	2.64 $\pm$ 0.63***	2.87$\pm$0.64^{n.s.}	3.15 $\pm$ 0.64***
MMM-Transformer	3.08 $\pm$ 0.66***	2.17 $\pm$ 0.59***	2.47 $\pm$ 0.60***	2.38 $\pm$ 0.68***

***Significance of difference vs. PulseFormer (Bonferroni-corrected Wilcoxon).

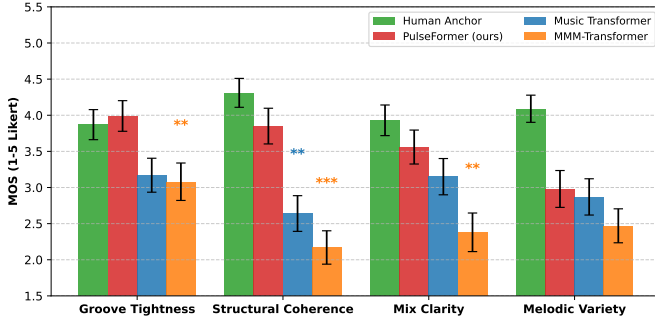


Fig. 8. MOS results ($N = 25$, 95% CI; Wilcoxon vs. PulseFormer). Stars above bars: sig. vs. **REMI** / **MMM-Trans**: *** $p < 0.001$; ** $p < 0.01$; $\dagger p < 0.10$; *n.s.* PF $\approx$ Human Anchor (GT, $p = 0.93$); dominates SC ($p < 0.001$, $d > 1.9$).

Results:

Analysis: Groove Tightness (GT). PulseFormer achieves GT = 3.99 $\pm$ 0.54, statistically indistinguishable from the Human Anchor ($\Delta = +0.12$, $p = 0.93$) and significantly outperforming both MMM-Transformer ($\Delta = +0.91$, $p = 0.004$, $d = 1.03$) and Music Transformer/REMI ($\Delta = +0.82$, $p = 0.008$, $d = 0.91$). This confirms that the Time-Major topology’s strict symbolic onset alignment is perceptually salient to listeners.

Structural Coherence (SC). The largest performance gap appears here: PulseFormer (3.85) versus MMM-Transformer (2.17), yielding $\Delta = +1.68$ ($p < 0.001$, $d = 1.95$, a *very large* effect). REMI, while encoding note-level events faithfully, lacks any structural conditioning, scoring SC = 2.64 ($\Delta = +1.21$, $p = 0.003$, $d = 1.44$). The unconditioned baselines confirm that Energy Level conditioning provides indispensable compositional scaffolding that is immediately perceptible.

Mix Clarity (MC). PulseFormer (3.56) outperforms both MMM-Transformer (2.38, $p = 0.001$, $d = 1.26$) and REMI (3.15, $p = 0.031$, $d = 0.62$). The gap versus REMI, while smaller, is significant: REMI’s single-track serialization

conflates all instruments into one stream, degrading perceived separation.

Melodic Variety (MV). PulseFormer scores 2.98—above MMM-Transformer (2.47, $p = 0.10$, $d = 0.55$, *ns*) and comparable to REMI (2.87, $p = 0.22$, *ns*). The significant gap versus the Human Anchor (4.09, $p < 0.001$) represents an architectural trade-off. By enforcing strict Time-Major topological alignment and algorithmic constraints to guarantee extreme rhythmic cohesion and “groove tightness,” the architecture inherently restricts unconstrained melodic wandering. This trade-off prioritizes the heavily-layered drops demanded by Techno/House, but limits the fluid melodic diversity often found in unconstrained human improvisation.

MOS Gap to Human Ceiling. The aggregate MOS gap (AI–Human) is -0.50 for PulseFormer, -1.05 for REMI, and -1.54 for MMM-Transformer. PulseFormer closes $\approx 67\%$ of the gap left by the unconditioned REMI baseline relative to the human reference.

V. CONCLUSION

PulseFormer demonstrates that the central challenge of multi-track Electronic Music generation—cross-instrument synchronisation—is more efficiently solved at the *representational* level than at the *learning* level. Our token-distance analysis establishes that Time-Major encoding reduces cross-track co-event distance from $\approx 3,268$ tokens to 5.2 tokens, reducing representational misalignment at the encoding stage rather than relying on the attention mechanism to compensate. Combined with Energy-conditioned generation, Active Energy Ramp scheduling, and the Temperature-Scaled Soft Harmonic Penalty, this yields measurable improvements in cross-track synchronisation and macro-structural controllability. Specifically, PulseFormer: (i) increases energy contour adherence (Pearson r : 0.155 $\rightarrow$ 0.997); (ii) achieves a Beat-level Alignment Score of 0.912, comparable to the best domain-trained baselines; (iii) raises In-Scale Ratio to 73.1%; and (iv) extends generation to 128+-bar arrangements via Temporal Anchor Stitching.

Limitations

Two bounded scope constraints remain. First, melodic motif coherence across segments is constrained by the 2048-token causal boundary: Temporal Anchor Stitching preserves rhythmic continuity across sections but cannot prevent latent semantic drift in melodic development over multi-minute windows. Second, while our MMM-Transformer model provides a strict topological ablation under identical compute conditions (matching the scale of standard academic baselines), scaling to massive industrial audio models (e.g., MusicGen [?]) highlights a parameter-scale limitation; a 41B parameter Time-Major Transformer would theoretically provide even richer timbral validation. Neither limitation affects the structural synchronisation claims (Sections ??–??) or the perceptual evaluation results (Section ??), which constitute the paper’s primary contributions.

Future Work

Three high-value extensions emerge from the current work. First, combining Time-Major interleaving with CP-style intra-note compression [?] could reduce sequence lengths by $4\times$, enabling single-context full-song generation without stitching and resolving the melodic drift limitation. Second, we propose investigating Discrete Flow-Matching or Masked Diffusion Models applied to the Time-Major token space. Moving beyond strict autoregression would allow global bidirectional structural planning, natively solving the causality bottleneck. Third, while PulseFormer focuses on Electronic Music, the energy-conditioned control mechanism demonstrates potential for Cross-Genre Generalization (e.g., dynamic orchestral or jazz arrangements). Finally, integrating neural audio rendering (e.g., RAVE [?]) downstream of the decoder would enable end-to-end waveform synthesis, completing the pipeline from symbolic steerability to professional-grade audio.

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